

Question

Different categories of employees are working in a software company like Regular Employees, Contract Employees, and Vendors. Their payroll differs. For Regular Employees: Basic Pay = 25000, HRA = 15000, TA = 5000. For Contract Employees: Basic Pay = 12000, TA = 3000, and no HRA. Write a Java program using **Interfaces** to display monthly salary details based on employee type.

Steps for Execution

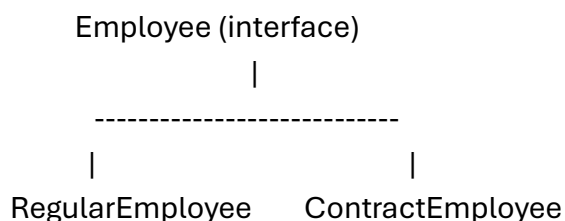
1. Open your Java IDE (e.g., Eclipse).
2. Create a file named EmployeeDemo.java.
3. Copy the program code into the file.
4. Save and compile using:

```
javac EmployeeDemo.java
```
5. Run the program:
6. Enter the Employee Id (e.g., R101 for Regular, C101 for Contract).
7. Observe the salary details displayed.

Algorithm

1. Define an Employee interface with method calculateSalary().
2. Create RegularEmployee and ContractEmployee classes implementing Employee.
3. Each class defines its own salary structure.
4. In EmployeeDemo main class:
 - Accept employee ID input.
 - Based on ID prefix (R or C), create the respective object.
 - Call calculateSalary() to display salary details.

UML Class Diagram



Code

```
import java.util.Scanner;
```

```
// Interface
```

```
interface Employee {  
    void calculateSalary();  
}
```

```
// Regular Employee class
```

```
class RegularEmployee implements Employee {  
  
    int basic = 25000;  
  
    int hra = 15000;  
  
    int ta = 5000;  
  
    public void calculateSalary() {  
  
        int total = basic + hra + ta;  
  
        System.out.println("Salary Details:");  
  
        System.out.println("Basic Pay: " + basic);  
  
        System.out.println("HRA: " + hra);  
  
        System.out.println("T.A: " + ta);  
  
        System.out.println("Total Amount: " + total);  
  
    }  
}
```

```
// Contract Employee class
```

```
class ContractEmployee implements Employee {  
  
    int basic = 12000;  
  
    int ta = 3000;
```

```
public void calculateSalary() {  
    int total = basic + ta;  
    System.out.println("Salary Details:");  
    System.out.println("Basic Pay: " + basic);  
    System.out.println("T.A: " + ta);  
    System.out.println("Total Amount: " + total);  
}  
}
```

```
public class EmployeeDemo {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter Employee Id: ");  
        String empld = sc.nextLine();  
  
        Employee emp;  
  
        if(empld.startsWith("R")) {  
            emp = new RegularEmployee();  
        } else if(empld.startsWith("C")) {  
            emp = new ContractEmployee();  
        } else {  
            System.out.println("Invalid Employee Id!");  
            sc.close();  
            return;  
        }  
    }  
}
```

```
emp.calculateSalary();  
sc.close();  
}  
}
```

Output Screenshot

Testcase 1:

Enter Employee Id: R101
Salary Details:
Basic Pay: 25000
HRA: 15000
T.A: 5000
Total Amount: 45000

Testcase 2:

Enter Employee Id: C101
Salary Details:
Basic Pay: 12000
T.A: 3000
Total Amount: 15000